COMPETENCY STANDARD

FOR

Industry Supervisor
(JUTE SECTOR)

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

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INTRODUCTION:

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

The Competency Standard also suggests integration of YouTube or similar platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standard has been reviewed and updated by a working group comprised of occupation-specific experts from the industry/institution, relevant consultants of SICIP.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Unit of Competency
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

Identification and validation of units of competency and elements for each occupation were made by experts from in consultative workshop.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.
- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a Glance:

Generic Competencies (20 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP- JUTE-IS-01-G	Apply occupational health and safety (OHS) practices at workplace	1. Identify, control and report OHS hazards 2. Conduct work safely 3. Follow emergency response procedures 4. Maintain and improve health and safety in the workplace	10
SICIP- JUTE-IS -02-G	Work In a Team Environment	1. Identify team goals and work processes 2. Communicate and co-operate with team members 3. Work as a team member 4. Solve problems as a team member	10
		Total Hour	**20**

Sector Specific Competencies (20 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP- JUTE-IS -01-S	Apply Quality Procedures in Jute Manufacturing	1. Identify quality procedures 2. Follow quality procedures 3. Maintain standard procedures	10
SICIP- JUTE-IS -02-S	Apply green practices in Jute sector	1. Interpret green concepts 2. Minimize resource use in the workplace 3. Implement waste management practices	10
		Total Hour	**20**

Occupation Specific Competencies (180 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP- JUTE-IS -01-O	Identify Jute Manufacturing Processes and Products	1. Interpret jute fiber properties 2. Identify jute manufacturing processes 3. Identify jute products and classification 4. Record materials requirements	20
SICIP- JUTE-IS -02-O	Supervise Production Operations in the Jute Mill	1. Interpret production schedules and plan daily production 2. Perform plant capacity calculation 3. Monitor machine operations 4. Supervise workers and the production 5. Maintain production records 6. Coordinate shift Plan	50
SICIP- JUTE-IS -03-O	Manage Quality Control in Jute Production process	1. Implement in-process quality checks 2. Identify jute product defects and deviations 3. Apply quality tools for jute production 4. Ensure conformance to buyer specifications	30
SICIP- JUTE-IS -04-O	Perform Inventory Management	1. Manage raw material inventory 5. Optimize material utilization	10
SICIP- JUTE-IS -05-O	Manage Workforce and Industrial Relations in the Jute Sector	1. Supervise and motivate workers 2. Manage attendance and performance 3. Address grievances and maintain workplace discipline 4. Comply with labor laws applicable to the jute industry	30
SICIP- JUTE-IS -06-O	Apply Process	1. Identify process	25

Code	Unit of Competency	Elements of Competency	Duration (hours)
	Improvement Techniques in Jute Manufacturing	inefficiencies 2. Apply lean manufacturing principles to jute production 3. Perform section-wise line balancing 4. Implement continuous improvement practices	
SICIP- JUTE-IS -07-O	Manage Personnel and Professional Development	1. Interpret personal development skills 2. Set and meet self-development priorities 5. Maintain professional growth and development	15
		Total Hours	**180**

COMPETENCY STANDARD: SWEATER LINKING OPERATION

Generic Competencies:

Unit of Competency: **APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICES AT WORKPLACE**	**Nominal Duration:** 10 Hrs**.**	**Unit Code:** SICIP-JUTE-IS-01-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to apply occupational health and safety (OHS) practices at workplace. It specifically includes the tasks of identifying, controlling and reporting OHS hazards, conducting work safely, following emergency response procedures and maintaining and improving health and safety in the workplace.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	**Performance Criteria**
1. Identify, control and report OHS hazards	1.1 Work area is routinely checked for Occupational Health and Safety (OHS) hazards prior to commencing and during work. 1.2 **<u>Hazards</u>** are identified and corrective action is taken within the level of responsibility. 1.3 OSH hazards and incidents are reported to appropriate personnel according to workplace procedures. 1.4 **<u>Safety signs and symbols</u>** are identified and followed. 1.5 **<u>Preventive Control measures</u>** are identified in accordance with OSH work standards.
2. Conduct work safely	2.1 OHS practices are applied in the workplace. 2.2 **<u>Personal Protective Equipment (PPE)</u>** are used.
3. Follow emergency response procedures	3.1 Emergency situations are identified and reported according to workplace requirements. 3.2 Emergency procedures are followed as appropriate to the nature of the emergency and according to workplace procedures. 3.3 **<u>Workplace procedures</u>** for dealing with accidents, fires and emergencies are followed whenever necessary within scope of responsibilities.
4. Maintain and improve health and safety in the workplace	4.1 Risks are identified and appropriate control measures are implemented in the workplace. 4.2 Recommendations arising from risk assessments are

	implemented within level of responsibility. 4.3 Safety records are maintained according to **<u>company policies</u>.**

Range of Variables

Variable	Range (May include but not limited to):
1. Hazards	1.1. OHS incidents include near misses, injuries, illnesses and property damage, noise, handling hazardous substances. 1.2. Biological hazards- bacteria, viruses, plants, parasites, mites, molds, fungi, insects 1.3. Chemical hazards – dusts, fibers, mists, fumes, smoke, gasses, vapors 1.4. Ergonomics Hazards 1.5. Physiological factors – monotony, personal relationship, work out cycle 1.6. Safety hazards (unsafe workplace condition) – confined space, excavations, falling objects, gas leaks, electrical, poor storage of materials and waste, spillage, waste and debris 1.7. Unsafe workers' act (Smoking in off-limited areas, Substance and alcohol abuse at work) 1.8. Working with and near moving equipment / load shifting equipment. 1.9. Broken or damaged equipment or materials.
2. Safety signs and symbols	2.1 Direction signs (exit, emergency exit, etc.) 2.2 First aid signs 2.3 Danger Tags 2.4 Hazard signs 2.5 Safety tags 2.6 Warning signs
3. Personal Protective Equipment (PPE)	3.1 Apron 3.2 Safety Helmet 3.3 Goggles 3.4 Ear muffs 3.5 Ear plugs 3.6 Gloves 3.7 Clothing 3.8 Safety Boots 3.9 Face Protection (Musk & Shield) 3.10 Scarf 3.11 Hair Net/cap/bonnet

	3.12 Hard hat
4. Preventive Control Measures	4.1 Eliminate the hazard (i.e., get rid of the dangerous machine 4.2 Isolate the hazard (i.e. keep the machine in a closed room and operate it remotely; barricade an unsafe area off) 4.3 Substitute the hazard with a safer alternative (i.e., replace the machine with a safer one) 4.4 Use administrative controls to reduce the risk (i.e. give trainings on how to use equipment safely; OHS-related topics, issue warning signage, rotation/shifting work schedule) 4.5 Use engineering controls to reduce the risk (i.e. use safety guards to machine) 4.6 Use personal protective equipment 4.7 Safety, health and work environment evaluation 4.8 Periodic and/or special medical examinations of workers
5. Workplace Procedures	5.1 Fire fighting 5.2 Medical and first aid 5.3 Evacuation 5.4 Material Safety Data Sheets (MSDSs) and manufacturers' advice
6. Company policies	6.1 Job related Standard Operating Procedures (SOPs). 6.2 Occupational Health and Safety (OHS) specific procedures. Examples of OHS procedures include – consultation and participation, emergency response to specific hazards, incident investigation, risk assessment, reporting arrangement and issue resolution procedures.

Curricular Content Guide

1. Underpinning Knowledge	1.1. OHS Workplace Policies and Procedures. 1.2. Work safety procedures. 1.3. Fire and emergency procedures. 1.4. Types of Hazards (Biological, Chemical and Physical) and their effects. 1.5. PPE types and uses. 1.6. Personal hygiene practices. 1.7. OHS awareness. 1.8. Steps of hazard identification. 1.9. Principles of hazards control. 1.10. Employer's role.

	1.11. Supervisor's responsibilities.
2. Underpinning Skills	2.1. Identifying OHS policies and procedures. 2.2. Following personal work safety practices. 2.3. Reporting hazards and risks. 2.4. Responding to emergency procedures. 2.5. Maintaining physical well-being in the workplace. 2.6. Identify tools and equipment related to OHS. 2.7. Improving OHS performance.
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties 3.4. Eagerness to learn 3.5. Tidiness and timeliness 3.6. Environmental concerns 3.7. Respect for rights of peers and seniors at workplace 3.8. Communication with peers and seniors at workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Workplace procedures 4.3 Manuals 4.4 Workplace documents, signs and symbols 4.5 Relevant specifications or work instructions. 4.6 Tools, equipment and facilities appropriate to the process or activities. 4.7 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 used Personal Protective Equipment (PPE). 1.2 identified hazards. 1.3 took corrective action of different hazards. 1.4 took corrective action for emergency procedure. 1.5 reported emergency situation to the Supervisor / Manager. 1.6 satisfied requirements mentioned in the performance criteria and range of variables.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated

	work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **WORK IN A TEAM ENVIRONMENT**	**Nominal Duration:**	**Unit Code:** SICIP-JUTE-IS-02-G

	10 Hrs.	

Unit Descriptor:
This unit covers the knowledge, skills and attitudes required to work in a team environment. It specifically includes the tasks of Identifying team goals and processes, communicating and cooperating with team members, working as a team member and solving problems as a team member.

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify team goals and work processes.	1.1. Team goals and collaborative decision-making processes are identified. 1.2. Role and common goals of the team are defined from available **<u>sources of information.</u>** 1.3. Team structure, responsibilities and reporting relations are identified from team discussions and other external sources.
2. Communicate and co-operate with team members.	2.1 Communication and negotiation skills are applied and maintained in all relevant situations. 2.2 Constructive contributions are made to **<u>workplace discussions</u>** on such issues as production, quality and safety. 2.3 Goals/ objectives and action plans undertaken in the workplace are communicated promptly. 2.4 Information regarding problems and issues are organized coherently to ensure clear and effective communication. 2.5 Dialogue is initiated with appropriate personnel. 2.6 Communication problems and issues are raised 2.7 Barriers to communication are identified and resolved
3. Work as a team member	3.1 Effective forms of communication are used to interact with **<u>team members</u>** in discussing team activities and objectives. 3.2 Mutual respect, empathy and active collaboration are demonstrated 3.3 Communication channels are followed as per **<u>workplace context.</u>**
4. Solve problems as a team member	4.1 Current and potential problems faced by team are identified 4.2 Problems are investigated and analyzed

	4.3 Potential solutions of problem are identified 4.4 Recommendations about possible solutions are developed, documented, ranked and presented to team members for decision.

Range of Variables

Variable	Range (May include but not limited to):
1. Sources of information	1.1. Organizational structures 1.2. Operations Manuals 1.3. Job description 1.4. Standard operating procedures
2. Workplace discussions	2.1 Coordination meetings 2.2 Toolbox discussion 2.3 Peer-to-peer discussion
3. Team members	3.1 Coach / members 3.2 Supervisor / manager 3.3 Peers / colleagues 3.4 Other members /Employee representative of the organization.
4. Workplace context	4.1 National laws and statutes 4.2 Standard operating procedures 4.3 Workplace rules and regulations

Curricular Content Guide

1. Underpinning Knowledge	1.1. Sources of information define 1.2. Team structure, role, and responsibility. 1.3. Individual member's roles and responsibilities. 1.4. Effective verbal communication methods 1.5. Communication flow and reporting structures. 1.6. Interpersonal communication skills. 1.7. Organization requirements for written and electronic communication methods 1.8. Communication problems and issues 1.9. Barriers in communication 1.10. Team planning. 1.11. Team meeting procedures. 1.12. Workplace etiquette 1.13. Industry maintenance, service and helpdesk practices, processes and procedures 1.14. Industry standard diagnostic tools 1.15. Malfunctions and resolutions
2. Underpinning Skills	2.1 Organizing sources of information 2.2 Identifying the role and responsibility of the team.

	2.3 Identifying roles and responsibilities of individual members. 2.4 Identifying effective verbal communication methods 2.5 Identifying communication flow and reporting structure. 2.6 Identifying interpersonal communication skills 2.7 Complying with organization requirements for the use of written and electronic communication methods 2.8 Negotiation and communication skills 2.9 Participating in team discussion. 2.10 Working as a team member. 2.11 Participating in a variety of workplace discussions 2.12 Effective clarifying and probing skills 2.13 Identifying issues 2.14 Identifying current industry standard diagnostic tools 2.15 Describing common malfunctions and resolutions. 2.16 Determining the root cause of a routine malfunction
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Eagerness to learn 3.5 Tidiness and timeliness 3.6 Environmental concerns 3.7 Respect for rights of peers and seniors at workplace 3.8 Communication with peers and seniors at workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Relevant specifications or work instructions. 4.4 Tools, equipment and facilities appropriate to the process or activities. 4.5 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified team goals and work processes 1.2 identified own role and responsibilities within team 1.1 communicated and cooperated with team members 1.2 practice problem solving within the team
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)

3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Sector Specific Competencies (20 Hrs.)

Unit of Competency: **APPLY QUALITY PROCEDURES IN JUTE MANUFACTURING**	**Nominal Duration:** 10 Hrs**.**	**Unit Code:** SICIP-JUTE-IS-01-S
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to apply quality procedures in jute manufacturing. It specifically includes the tasks of identifying quality procedures, following quality procedures and maintaining standard procedures.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	**Performance Criteria**
1. Identify quality procedures	1.1. **Manuals** and guidelines for jute production processes are collected as per product specifications. 1.2. The importance of manuals in ensuring quality standards is recognized. 1.3. Instructions and procedures for quality control are identified. 1.4. Required information is collected from manuals. 1.5. Performance measurement systems are identified.
2. Follow quality procedures	2.1 Instructions and procedures are followed strictly and duties are performed in accordance with the **quality improvement system**. 2.2 Concept of supplying jute products to meet **customer quality requirements** is understood and applied. 2.3 Conformance to specifications is ensured. 2.4 Defects and deviations are detected and reported to the relevant authority according to standard operating procedures.
3. Maintain standard procedures	3.1 Performance is assessed at regular intervals. 3.2 Specifications and standard operating procedures are established. 3.3 Quality of jute product is checked and verified. 3.4 Quality control and quality assurance system procedures for each job are followed. 3.5 Conformance to specification is ensured in all situations.

Range of Variables

Variable	Range (May include but not limited to):
1. Manuals	1.1 Buyers' specification manual 1.2 Compliance manual 1.3 Jute product quality manual 1.4 Maintenance procedure manual 1.5 Signs and symbols, instruction manuals
2. Quality improvement system	2.1 Quality inspection 2.2 Testing 2.3 Quality control 2.4 Quality assurance 2.5 Total Quality Management (TQM)
3. Customer quality requirements	3.1 Performance 3.2 Features 3.3 Reliability 3.4 Conformance 3.5 Aesthetics 3.6 Durability

Curricular Content Guide

1. Underpinning Knowledge	1.1 Themes on various types of RMG manuals 1.2 Schedules, dimensions and specifications contained in sketch 1.3 Units of conversion 1.4 Rules of sketch, drawings and specifications 1.5 Sketch and specifications 1.6 Signs and symbols 1.7 Terms and abbreviations used
2. Underpinning Skills	2.1 Recognising importance of manual 2.2 Selecting appropriate manuals as per sample 2.3 Collecting information from manual as per sample 2.4 Identifying signs and symbols 2.5 Identifying drawings and specifications 2.6 Interpreting schedules, dimensions, drawings and specifications
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in workplace 3.8 Communication with peers, sub-ordinates and seniors in

	workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Manuals 4.3 Workplace documents, signs and symbols 4.4 Sketch and specifications 4.5 Tools, equipment and facilities appropriate to the process or activities. 4.6 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 collected information from manual as per sample 1.2 Identified relevant drawings and specifications 1.3 identified signs and symbols 1.4 identified terms and abbreviations 1.5 identified sketches and specifications as per sample
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **APPLY GREEN PRACTICES IN JUTE SECTOR**	**Nominal Duration:** 10 hrs.	**Unit Code:** SICIP-JUTE-IS-02-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required for apply green practices in Jute sector. It specifically includes the tasks of interpreting green concepts, minimizing resources use in the workplace and implementing waste management practices.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret green concepts	1.1 **Principles of green practices in Jute** manufacturing are explained. 1.2 Key terms and symbols in Jute production processes are identified. 1.3 **Sources of environmental impacts** during Jute manufacturing activities are identified. 1.4 Jute manufacturing activities use are explained. 1.5 **Ways of mitigating environmental impacts** in Jute manufacturing are explained.
2. Minimize resources use in the workplace	2.1 Water, energy, and raw material consumption in Jute manufacturing are documented. 2.2 **Recyclable and non-recyclable items** in Jute production processes are identified. 2.3 Procedures to reduce resource consumption in Jute manufacturing are implemented. 2.4 Sustainable alternatives in Jute manufacturing are explored and applied.
3. Implement waste management practices	3.1 **Different types of waste in Jute manufacturing** are identified. 3.2 Hazardous waste in jute production is disposed of according to environmental regulations. 3.3 Green habits to reduce waste in personal and professional life are practiced.

Range of Variables

Variable	Range (May include but not limited to):
1. Principles of green practices in RMG (Ready-Made Garment)	1.1 Reducing energy consumption in Jute manufacturing processes.

	1.2 6R for waste management in Jute production: 1.2.1 Refuse unnecessary materials and processes. 1.2.2 Reduce resource consumption and waste generation. 1.2.3 Reuse materials within the production process. 1.2.4 Recycle materials for future use in Jute manufacturing. 1.2.5 Recover usable resources from waste. 1.2.6 Repair damaged products or materials instead of discarding them. 1.3 Use of sustainable materials in Jute production with low environmental impact. 1.4 Recycling of materials used in the Jute manufacturing process. 1.5 Sustainable transportation methods for moving goods and materials within the Jute sector.
2. Sources of environmental impacts	2.1 Air pollution in Jute manufacturing 2.2 Water pollution in Jute manufacturing 2.3 Soil degradation in Jute manufacturing 2.4 Noise pollution in Jute manufacturing 2.5 Waste generation in Jute manufacturing 2.6 Resource depletion in Jute manufacturing
3. Ways of mitigating environmental impacts	3.1 Utilizing energy-efficient equipment in Jute production processes. 3.2 Adopting renewable energy sources for Jute manufacturing operations. 3.3 Implementing site protection measures to prevent pollution and reduce environmental. 3.4 Using reusable materials in production processes. 3.5 Choosing sustainable materials with low environmental impact for garment production. 3.6 Using noise-reducing equipment and scheduling production work. 3.7 Optimizing logistics and delivery to reduce transportation emissions and improve overall supply chain efficiency.
4. Recyclable and non-recyclable items	4.1 **Recyclable Items in RMG Manufacturing** 4.1.1 Metal components. 4.1.2 Wooden materials. 4.1.3 Fabric scraps and textile fibers. 4.1.4 Plastic items, including buttons, tags, and packaging materials. 4.2 **Non-Recyclable Items in Jute Manufacturing** 4.2.1 Paints, coatings, and dyes. 4.2.2 Contaminated materials.

	4.2.3 Treated wood and other materials.
5. Different types of waste in RMG manufacturing	5.1 Fabric waste from cutting and trimming processes. 5.2 Wood waste from packaging or display materials. 5.3 Metal waste from trims, buttons, zippers, and other hardware. 5.4 Textile waste, including scraps from garment production and defective products. 5.5 Plastic waste, such as packaging materials, labels, and plastic buttons. 5.6 Solid waste from chemical treatments or dyeing processes. 5.7 Packaging waste from raw materials and finished products.

Curricular Content Guide

1. Underpinning Knowledge	1.1 Principles of green practices in Jute manufacturing. 1.2 Sources of environmental impacts. 1.3 Ways of mitigating environmental impacts 1.4 Recyclable and non-recyclable items 1.5 Different types of waste in Jute manufacturing 1.6 Hazardous waste in Jute production 1.7 Green habits to reduce waste
2. Underpinning Skills	2.1 Explaining principles of green practices in Jute manufacturing 2.2 Identifying sources of environmental impacts 2.3 Explaining re-cyclable and non-recyclable items 2.4 Minimizing resources use in the workplace 2.5 Identifying different types of waste in Jute manufacturing 2.6 Disposing hazardous waste in Jute production 2.7 Practicing green habits to reduce waste
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Workplace documents, signs and symbols

	4.4 Codes of conduct 4.5 Tools, equipment and facilities appropriate to the process or activities. 4.6 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 interpreted green concepts 1.2 minimized resources use in the workplace 1.3 implemented waste management practices
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Occupation Specific Competencies (180 Hrs.)

Unit of Competency: **IDENTIFY JUTE MANUFACTURING PROCESSES AND PRODUCTS**	**Nominal Duration:** 20 Hrs**.**	**Unit Code:** SICIP-JUTE-IS-01-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to identify jute manufacturing processes and products. It specifically includes the tasks of interpreting jute fiber properties, identifying jute manufacturing processes, identifying jute products and classifying and recording materials requirements.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret jute fiber properties	1.1 **Jute fiber properties and grades** are identified and explained. 1.2 Fiber quality parameters are described and assessed. 1.3 Raw jute grading standards are interpreted as per industry norms. 1.4 Moisture content and fiber strength criteria are identified.
2. Identify jute manufacturing processes	2. 1 Types of **jute manufacturing processes** are identified. 2. 2 The sequence of jute manufacturing operations (from raw jute to finished product) is explained. 2. 3 Machinery and equipment for each manufacturing stage are identified. 2. 4 Functions of key jute processing machines are described.
3. Identify jute products and classification	3. 1 Different types of **jute products** are identified. 3. 2 Classification of jute products based on end use is described. 3. 3 Jute product specifications and standards are interpreted as per buyer/market requirements. 3. 4 Value-added jute products are identified
4. Record materials requirements	4. 1 Different types of materials used in jute production are identified. 4. 2 **Bill of materials (BOM)** for jute products is identified. 4. 3 Materials are itemized and recorded as per production requirements.

Range of Variables

Variable	Range (May include but not limited to):
1. Jute fiber properties	1.1 W1 (Best quality) 1.2 W2 1.3 W3 1.4 BTR (Better quality) 1.5 CS (Cross) 1.6 Rejections 1.7 Cuttings and carding waste
2. Jute grades	**Tossa** 2.1 KACCHA (Raw): Top, Grade-A, Grade-B, Grade-C, Cross, SMR 2.2 PACCA (Processed): Special, Grade-A, Grade-B, Grade-C, Grade-D, Grade-E. **White (Deshi)** 2.3 KACCHA (Raw): Top, Grade-A, Grade-B, Grade-C, Cross, SMR 2.4 PACCA (Processed): Special, Grade-A, Grade-B, Grade-C, Grade-D, Grade-E **Mesta** 2.5 KACCHA (Raw): Top, Grade-A, Grade-B, Grade-C. 2.6 PACCA (Processed): Special, Grade-A, Grade-B, Grade-C. **Kenaf** 2.7 KACCHA (Raw): Top, Grade-A, Grade-B, Grade-C 2.8 PACCA (Processed): Special, Grade-A, Grade-B, Grade-C
3. Jute manufacturing processes	3.1 Batching and softening 3.2 Carding (breaker and finisher) 3.3 Drawing (1st, 2nd, 3rd) 3.4 Spinning (ring frame) 3.5 Winding and beaming 3.6 Weaving (plain, CBC (Carpet backing cloth), twill) 3.7 Finishing (calendaring, Mangling, cutting, stitching) 3.8 Packing
4. Jute products	4.1 Hessian cloth (burlap) 4.2 Sacking and bags 4.3 Carpet backing cloth (CBC) 4.4 Jute yarn and twine 4.5 Geo-textiles 4.6 Jute composite products 4.7 Jute handicrafts and diversified products 4.8 Jute-cotton blended textiles
5. Bill of materials (BOM)	5.1 Item description 5.2 Material specifications

	5.3 Quantity required 5.4 Unit of measurement 5.5 Raw jute grade 5.6 Chemicals and sizing agents 5.7 Packaging materials

Curricular Content Guide

1. Underpinning Knowledge	1.1 Jute fiber properties, grades, and quality parameters 1.2 Jute manufacturing process flow from retting to finished product 1.3 Key machinery and equipment in jute manufacturing 1.4 Types and classification of jute products 1.5 Jute product specifications and market standards 1.6 Bill of materials preparation for jute production 1.7 Value-added and diversified jute products
2. Underpinning Skills	2.1 Identifying and explaining jute fiber properties 2.2 Identifying jute manufacturing processes 2.3 Identifying and classifying jute products 2.4 Recording materials as per bill of materials (BOM) 2.5 Interpreting product specifications and standards
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Workplace documents, signs and symbols 4.4 Tools, equipment and facilities appropriate to the process or activities. 4.5 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 interpreting jute fiber properties 1.2 identifying jute manufacturing processes 1.3 identify jute products and classification

	1.4 recording materials requirements.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency:	Nominal	Unit Code:

SUPERVISE PRODUCTION OPERATIONS IN THE JUTE MILL	**Duration:** 50 Hrs**.**	SICIP-JUTE-IS-02-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to supervise production operations in the jute mill. It specifically includes the tasks of interpreting production schedules and plan daily production, performing plant capacity calculation, supervising workers on the production, maintaining production records and coordinating shift plan.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **<u>bold and underlined</u>** are elaborated in the range of variables).

Elements of Competency	**Performance Criteria**
1. Interpret production schedules and plan daily production	1. 1 **<u>Time and action (TNA) plan</u>** is interpreted on basis of delivery lead time. 1. 2 Production schedules are interpreted and broken down into daily/weekly targets considering critical milestones and risk factors. 1. 3 Daily production targets are interpreted from **<u>production schedules.</u>** 1. 4 Work assignments are allocated to workers based on skill and machine availability. 1. 5 Production priorities are set based on delivery schedules and buyer requirements according to raw material availability 1. 6 Adjustments to production schedules due to operational constraints are communicated to management.
2. Perform plant capacity calculation	2.1 **<u>Plant capacity calculation formula</u>** for jute production is identified. 2.2 Plant capacity calculation is carried out for spinning, weaving, and finishing sections. 2.3 Capacity vs. order requirements is compared and gaps are identified. 2.4 Capacity utilization is monitored and reported
3. Monitor machine operations	3. 1 Key **<u>production machines</u>** are identified and their operating parameters are verified. 3. 2 Machine performance indicators are monitored during production. 3. 3 Minor machine issues are identified and reported to maintenance in time. 3. 4 Machine downtime is recorded and analyzed for improvement.
4. Supervise workers and the production	4. 1 Workers are briefed on production targets and quality standards at the start of each shift.

	4. 2 Worker performance is observed and feedback is provided constructively. 4. 3 Non-conforming work practices are identified and corrected. 4. 4 Worker attendance and deployment records are maintained.
5. Maintain production records	5. 1 Daily production output is recorded accurately per section/machine. 5. 2 Wastage and downtime records are maintained. 5. 3 Production report are compiled and submitted to supervisors/management. 5. 4 Discrepancies in **<u>production records</u>** are identified and resolved.
6. Coordinate shift plan	6. 1 **<u>Shift handover and takeover</u>** procedures are followed as per organizational SOPs. 6. 2 Pending work, machine status, and issues are communicated clearly to the incoming supervisor. 6. 3 Handover and takeover records are completed and signed off.

Range of Variables

Variable	**Range** (May include but not limited to):
1. Time and action (TNA) plan	1.1 Lead time 1.2 Raw material procurement 1.3 Production start date 1.4 Quality inspection checkpoints 1.5 Packing and dispatch 1.6 Shipment date
2. Production schedule	2. 1 Machine capacity 2. 2 Available workforce 2. 3 Raw material stock 2. 4 Buyer delivery schedule 2. 5 Machine maintenance schedule 2. 6 Quality requirements
3. Plant capacity calculation factors	3. 1 Total number of machines (by section) 3. 2 Total working hours per day 3. 3 Total number of workers per shift 3. 4 Standard machine output per hour 3. 5 Efficiency percentage
4. Production machines	4. 1 Carding machine (breaker and finisher card)

	4. 2 Drawing frames (1st, 2nd, 3rd, 4th) 4. 3 Spinning frames 4. 4 Roll Winding and reeling machines 4. 5 Cop winding machine 4. 6 Looms (hessian, CBC, sacking) 4. 7 Calendaring, Mangling machine 4. 8 Cutting and stitching machines
5. Production records	5. 1 Daily production report 5. 2 Machine downtime report 5. 3 Wastage report 5. 4 Worker attendance and deployment record 5. 5 Shift handover and takeover log 5. 6 Quality inspection report
6. Shift handover and takeover	6. 1 Production report 6. 2 Machine status report 6. 3 Quality issues 6. 4 Maintenance requests 6. 5 Safety incidents 6. 6 Worker issues

Curricular Content Guide

1. Underpinning Knowledge	1.1 Production planning and scheduling principles 1.2 Types and functions of jute processing machinery 1.3 Machine monitoring and performance indicators 1.4 Worker supervision principles and techniques 1.5 Production record keeping and documentation 1.6 Shift management and handover procedures 1.7 OHS compliance in jute production supervision
2. Underpinning Skills	2. 1 Planning daily production activities 2. 2 Monitoring machine operations and performance 2. 3 Supervising workers effectively on the production floor 2. 4 Maintaining accurate production records 2. 5 Coordinating shift handover effectively 2. 6 Communicating production targets and quality standards to workers
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace

4. Resource Implications	The following resources must be provided: 4. 1 Workplace (simulated or actual) 4. 2 Standard operating procedure 4. 3 Workplace documents, signs and symbols 4. 4 Tools, equipment and facilities appropriate to the process or activities. 4. 5 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Planned daily production activities 1.2 Monitored machine operations 1.3 Supervised workers on the production floor 1.4 Maintained production records 1.5 Coordinated shift handover and takeover
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **MANAGE QUALITY CONTROL IN JUTE**	**Nominal Duration:**	**Unit Code:** SICIP-JUTE-IS-03-O

PRODUCTION PROCESS	30 Hrs.	
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to manage quality control in jute production process. It specifically includes the tasks of implementing in-process quality checks, identifying jute product defects and deviations, applying quality tools for jute production and ensuring conformance to buyer specifications.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Implement in-process quality checks	1.1 **In-process quality parameters** for each production stage are identified. 1.2 Quality check procedures are implemented at designated inspection points. 1.3 Quality check results are recorded in the appropriate format. 1.4 Non-conforming materials are identified and segregated according to procedures
2. Identify jute product defects and deviations	2.1. Types of **jute product defects** are identified as per product category. 2.2. Defects are inspected and recorded including type and location. 2.3. Defective products are segregated from conforming products. 2.4. Root causes of common defects are identified.
3. Apply quality tools for jute production	3. 1 **Basic quality tools** are selected according to the quality problem. 3. 2 Quality tools are used to analyze defect patterns and production data. 3. 3 Results of quality tool application are documented. 3. 4 Corrective actions based on quality tool analysis are recommended and implemented.
4. Ensure conformance to buyer specifications	4. 1 **Buyer specifications** for jute products are interpreted accurately. 4. 2 Conformance to specifications is verified at each production stage. 4. 3 Non-conforming products are reported to management with documented evidence. 4. 4 Non-conforming products are segregated as per SOP. 4. 5 Corrective and preventive actions are followed up.

Range of Variables

Variable	Range (May include but not limited to):
1. In-process quality parameters	1.1 Yarn count/lea strength 1.2 Fabric weight (GSM) 1.3 Fabric width and weave density 1.4 Fabric tensile and burst strength 1.5 Thread density (EPI/PPI) 1.6 Moisture content of yarn and fabric
2. Jute product defects	2.1 **Yarn defects:** thick/thin places, slubs, weak yarn, uneven twist 2.2 **Fabric defects:** missing picks, broken ends, float, reed marks, weaving faults 2.3 **Finishing defects:** uneven width, edge defects, stains, holes 2.4 **Bag/sacking defects**: poor stitching, size variation, print misalignment
3. Basic quality tools	3. 1 Check sheet 3. 2 Control chart 3. 3 Histogram 3. 4 Pareto chart 3. 5 Scatter diagram 3. 6 Flow chart
4. Buyer specifications	4. 1 Weight/GSM 4. 2 Width and length 4. 3 Weave construction 4. 4 Tensile/burst strength 4. 5 Moisture regain 4. 6 Marking and labelling 4. 7 Packing requirements

Curricular Content Guide

1. Underpinning Knowledge	1.1 Quality control principles applied to jute manufacturing 1.2 In-process quality parameters and inspection points 1.3 Types and causes of jute product defects 1.4 Basic quality tools and their application 1.5 Buyer specification interpretation and conformance verification 1.6 Documentation and record-keeping for quality control 1.7 Corrective and preventive action procedures
2. Underpinning Skills	2.1 Implementing in-process quality checks 2.2 Identifying jute product defects 2.3 Applying quality tools for defect analysis 2.4 Ensuring conformance to buyer specifications 2.5 Recording and reporting quality data

	2.6 Recommending corrective actions
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Jute product samples (conforming and defective) 4.3 Quality check sheets and checklists 4.4 Basic quality tool templates (check sheets, charts) 4.5 Buyer specification documents (samples) 4.6 Tools, equipment and facilities appropriate to the process or activities.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Implemented in-process quality checks 1.2 Identified jute product defects 1.3 Applied quality tools for analysis 1.4 Ensured conformance to buyer specifications
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **PERFORM INVENTORY MANAGEMENT**	**Nominal Duration:** 10 Hrs.	**Unit Code:** SICIP-JUTE-IS-04-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform inventory management. It specifically includes the tasks of managing raw material inventory, and optimizing material utilization.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Manage raw material inventory	1.1 **Types of inventories** in jute manufacturing are identified. 1.2 Minimum and maximum stock levels for raw jute and other materials are set. 1.3 Inventory levels are monitored and restocking requests are initiated. 1.4 Inventory records are maintained accurately.
2. Optimize material utilization	2.1 Efficiency of raw jute and material consumption is identified. 2.2 Material wastage in jute production processes is measured and recorded. 2.3 **Material utilization percentage** is calculated as per production data. 2.4 Actions to reduce material wastage are identified and implemented.

Range of Variables

Variable	Range (May include but not limited to):
1. Types of inventories	1.1 Raw jute (by grade) 1.2 Chemicals and sizing agents 1.3 Packaging materials (bags, bags, baling) 1.4 Spare parts and consumables 1.5 Work-in-process (WIP) 1.6 Finished goods
2. Material utilization factors	2.1 Raw jute input weight 2.2 Finished product weight 2.3 Waste generated at each stage 2.4 Extraction/yield percentage 2.5 Carding, spinning and weaving losses

Curricular Content Guide

1. Underpinning Knowledge	1.1 Types of inventories in jute manufacturing 1.2 Inventory management principles and minimum/maximum stock levels 1.3 Material utilization and wastage calculation 1.4 Material requirement planning for jute production
2. Underpinning Skills	2.1 Managing raw material inventory 2.2 Calculating and optimizing material utilization 2.3 Maintaining inventory records 2.4 Communicating capacity and scheduling issues
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Production schedule templates 4.3 TNA plan samples 4.4 Inventory record forms 4.5 Tools, equipment and facilities appropriate to the process or activities. 4.6 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 managed raw material inventory 1.2 optimized material utilization
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **MANAGE WORKFORCE AND INDUSTRIAL RELATIONS IN THE JUTE SECTOR**	**Nominal Duration:** 30 Hrs**.**	**Unit Code:** SICIP-JUTE-IS-05-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to manage workforce and industrial relations in the jute sector. It specifically includes the tasks of supervising and motivating workers, managing attendance and performance, addressing grievances and maintaining workplace discipline and complying with labor laws applicable to the jute industry.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **<u>bold and underlined</u>** are elaborated in the range of variables).

Elements of Competency	**Performance Criteria**
1. Supervise and motivate workers	1.1 Work instructions are communicated clearly to production workers. 1.2 Workers are supervised to ensure adherence to production and quality standards. 1.3 **<u>Motivation techniques</u>** appropriate to the jute sector workforce are applied. 1.4 Individual worker strengths are identified and optimally utilized in production assignments.
2. Manage attendance and performance	2.1 Worker attendance is recorded and managed as per organizational procedures. 2.2 **<u>Worker performance</u>** is evaluated against set production targets. 2.3 Performance feedback is provided constructively and timely. 2.4 Underperforming workers are identified and remedial support is organized.
3. Address grievances and maintain workplace discipline	3.1 Worker grievances are listened to respectfully and recorded. 3.2 **<u>Grievance handling procedures</u>** are followed as per company policy. 3.3 Disciplinary issues are addressed using established disciplinary procedures. 3.4 A positive and respectful workplace culture is promoted.
4. Comply with labor laws applicable to the jute industry	4.1 Key provisions of the Bangladesh **<u>Labor Act relevant to jute sector</u>** workers are identified. 4.2 Workers' rights including wages, working hours, and leave entitlements are recognized. 4.3 Legal compliance in attendance, wages, and safety

	matters is maintained. 4.4 Child labor and forced labor prohibition provisions are correctly applied.

Range of Variables

Variable	Range (May include but not limited to):
1. Motivation techniques	1.1 Recognition and appreciation 1.2 Fair and transparent task allocation 1.3 Skill development opportunities 1.4 Team-based incentives 1.5 Regular feedback and communication
2. Worker's performance parameters	2.1 Daily production output vs target 2.2 Quality of output 2.3 Attendance and punctuality 2.4 Machine/equipment care 2.5 Cooperation and teamwork
3. Grievance handling procedures	3.1 Verbal warning 3.2 Written warning 3.3 Formal grievance filing 3.4 Mediation and resolution 3.5 Escalation to management or HR
4. Labor Act relevant to jute sector	4.1 Bangladesh Labor Act 2006 (as amended) 4.2 Working hours and overtime regulations 4.3 Minimum wage for jute sector 4.4 Annual leave and festival leave 4.5 Maternity benefits and gender responsive support 4.6 Child labor prohibitions 4.7 Safety and health obligations of employer

Curricular Content Guide

1. Underpinning Knowledge	1.1 Principles of workforce supervision in manufacturing environments 1.2 Motivation techniques applicable to jute sector workers 1.3 Attendance and performance management procedures 1.4 Grievance handling and disciplinary procedures 1.5 Bangladesh Labor Act provisions relevant to jute sector 1.6 Workers' rights and employer obligations 1.7 Child labor and forced labor prohibition
2. Underpinning Skills	2.1 Communicating work instructions clearly 2.2 Supervising and motivating workers 2.3 Managing attendance and evaluating worker performance

	2.4 Handling grievances and maintaining workplace discipline 2.5 Applying relevant labor law provisions 2.6 Maintaining accurate workforce records
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Attendance and performance record templates 4.3 Grievance handling procedure documents 4.4 Bangladesh Labor Act reference materials 4.5 Case study scenarios for grievance handling 4.6 Tools, equipment and facilities appropriate to the process or activities. 4.7 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Supervised and motivated workers effectively 1.2 Managed attendance and performance 1.3 Addressed grievances and maintained workplace discipline 1.4 Complied with labor laws applicable to the jute industry
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency:	**Nominal**	**Unit Code:**

APPLY PROCESS IMPROVEMENT TECHNIQUES IN JUTE MANUFACTURING	**Duration:** 25 Hrs**.**	SICIP-JUTE-IS-06-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to apply process improvement techniques in jute manufacturing. It specifically includes the tasks of identifying process inefficiencies, applying lean manufacturing principles to jute production, performing section-wise balancing, and implementing continuous improvement practices.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify process inefficiencies	1. 1 Bottleneck operations in the jute manufacturing process are identified. 1. 2 **<u>Types of waste in jute manufacturing</u>** are identified using lean principles. 1. 3 Root causes of process inefficiencies are analyzed. 1. 4 Potential areas for improvement are prioritized based on impact and feasibility.
2. Apply lean manufacturing principles to jute production	2. 1 Lean manufacturing concepts are explained and applied to jute production context. 2. 2 Value and non-value adding activities in jute manufacturing are identified. 2. 3 **<u>Lean tools applicable to the jute sector</u>** are identified and applied. 2. 4 Impact of lean implementation on productivity and quality is described.
3. Perform section-wise line balancing	3. 1 Production output per section (carding, drawing, spinning, weaving) is assessed. 3. 2 Capacity of each section is compared and imbalances are identified. 3. 3 **<u>Line balancing techniques</u>** are applied to reduce idle time and maximize output. 3. 4 Balancing loss is calculated and reported.
4. Implement continuous improvement practices	4. 1 **<u>Continuous improvement culture (Kaizen)</u>** is promoted in the workplace. 4. 2 Small improvement suggestions from workers are collected and evaluated. 4. 3 Implemented improvements are documented and monitored for effectiveness. 4. 4 Results of improvement actions are communicated to management.

Range of Variables

Variable	Range (May include but not limited to):
1. Types of waste in jute manufacturing (LEAN)	1. 1 Overproduction of yarn and fabric 1. 2 Excess inventory/WIP between sections 1. 3 Unnecessary machine downtime (waiting) 1. 4 Excess transportation of materials 1. 5 Over-processing (unnecessary re-carding) 1. 6 Defects and rework 1. 7 Unused worker skills and knowledge
2. Lean tools applicable to jute manufacturing	2. 1 5S (Sort, Set in order, Shine, Standardize, Sustain) 2. 2 Kaizen (continuous improvement events) 2. 3 Value Stream Mapping (VSM) 2. 4 Visual management and signboards 2. 5 Standardized work procedures 2. 6 Root cause analysis (5 Whys) 2. 7 TPM (Total Productive Maintenance) 2. 8 Kanban for material flow control
3. Line balancing techniques	3. 1 Split task across machines 3. 2 Share task between workers 3. 3 Use parallel workstations 3. 4 Improve material supply to bottleneck sections 3. 5 Adjust workforce deployment 3. 6 Incentive-based motivation
4. Continuous improvement (Kaizen) activities	4. 1 Daily 5-minute improvement meetings 4. 2 Suggestion box system 4. 3 Small-group improvement activities 4. 4 Cross-functional improvement teams 4. 5 Monthly improvement review meetings

Curricular Content Guide

1. Underpinning Knowledge	1. 1 Lean manufacturing principles and their application to jute sector 1. 2 Types of waste in jute manufacturing (TIM WOODS / MUDA) 1. 3 Value stream mapping for jute production 1. 4 Section-wise capacity assessment and line balancing 1. 5 Balancing loss calculation 1. 6 Kaizen and continuous improvement methodologies 1. 7 Root cause analysis techniques
2. Underpinning Skills	2. 1 Identifying process inefficiencies and bottlenecks 2. 2 Applying lean tools to jute manufacturing context 2. 3 Performing section-wise line balancing 2. 4 Calculating and reducing balancing loss 2. 5 Implementing and documenting continuous improvement activities 2. 6 Facilitating worker participation in improvement
3. Underpinning Attitudes	3. 1 Commitment to occupational health and safety 3. 2 Promptness in carrying out activities

	3. 3 Sincere and honest to duties 3. 4 Environmental concerns 3. 5 Eagerness to learn 3. 6 Tidiness and timeliness 3. 7 Respect for rights of peers and seniors in the workplace 3. 8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4. 1 Workplace (simulated or actual) 4. 2 Production data sheets 4. 3 Value stream mapping templates 4. 4 Line balancing worksheets 4. 5 Tools, equipment and facilities appropriate to the process or activities. 4. 6 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1. 1 Identified process inefficiencies 1. 2 Applied lean manufacturing principles to jute production 1. 3 Performed section-wise line balancing 1. 4 Implemented continuous improvement practices
2. Methods of Assessment	Competency should be assessed by: 2. 1 Written test 2. 2 Practical demonstration 2. 3 Oral questioning 2. 4 Portfolio (Optional)
3. Context of Assessment	3. 1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3. 2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **MANAGE PERSONNEL AND PROFESSIONAL DEVELOPMENT**	**Nominal Duration:** 15 Hrs.	**Unit Code:** SICIP-JUTE-IS-07-0
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to manage personnel and professional development. It specifically includes the tasks of interpreting personnel development skills, setting and meet self-development priorities and maintaining professional growth and development.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret personnel development skills	1.1 Objectives of personnel development skills are described. 1.2 **Personnel development** skills are identified. 1.3 Intra and interpersonal relationships are maintained. 1.4 Self-analysis is performed and personal development needs are identified.
2. Set and meet self-development priorities	2.1 Tasks are prioritized to achieve personal, team and organizational goals. 2.2 **Resources** are utilized efficiently and effectively to manage work priorities. 2.3 Economic usage and maintenance of facilities are followed as per established procedures.
3. Maintain professional growth and development	3.1 Pro-activeness is demonstrated in fulfilling personal and professional growth requirements. 3.2 **Training and career opportunities** are identified and accessed based on job requirements. 3.3 **Recognitions** are sought and demonstrated as proof of career advancement. 3.4 **Certifications** relevant to the jute sector are obtained and renewed.

Range of Variables

Variable	Range (May include but not limited to):
1. Personnel development skills	1.1 Problem-solving 1.2 Self-confidence 1.3 Adaptability 1.4 Integrity

	1.5 Work ethic 1.6 Pro-activeness 1.7 Leadership
2. Resources	2.1 Human 2.2 Financial 2.3 Technology
3. Training and career opportunities	3.1 Participation in jute sector training programs 3.2 Technical and supervisory training 3.3 Managerial training 3.4 Serving as resource persons in workshops
4. Recognitions	4.1 Recommendations 4.2 Citations 4.3 Certificates of Appreciation 4.4 Commendations 4.5 Awards and tangible rewards
5. Certifications	5. 1 National Certificates 5. 2 Certificate of Competency 5. 3 Industry-specific supervisor credentials

Curricular Content Guide

1. Underpinning Knowledge	1.1 Workplace communication policies, standards and procedures. 1.2 Verbal and non-verbal communication. 1.3 Modes of communication
2. Underpinning Skills	2.1 Receiving verbal instructions 2.2 Interpreting verbal and written information/ instruction 2.3 Conveying instructions using verbal and 2.4 Written forms of communication 2.5 Participating in work place meetings and discussion.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Eagerness to learn 3.5 Tidiness and timeliness 3.6 Environmental concerns 3.7 Respect for rights of peers and seniors at workplace 3.8 Communication with peers and seniors at workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Writing materials. 4.3 Workplace documents, signs and symbols 4.4 Tools, equipment and facilities appropriate to the process or activities. 4.5 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.3 interpreted verbal Instructions. 1.4 interpreted written instructions are read. 1.5 followed routine written instructions in sequence. 1.6 used proper communication methods to transmit instructions.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

End of the Competency Standard

Workshop/Lab Facility Standard

Course Name:	**Industry Supervisor**
Number of Trainees:	30

Course-wise Training Space (Theoretical Classroom, Workshop/ Lab/ Classroom cum Workshop):

- Classroom – 350 sft (33 sqm)
- Workshop/ lab – 800 sft (75 sqm)

  OR
- Classroom cum workshop – 1000 sft (93 sqm)

Major Training Equipment and Training Facilities:

S.N.	Major Equipment and Training facilities	Required facilities
1.	Computer/Laptops	01
2.	Multimedia projector & screen	01
3.	Internet connectivity	01
4.	Whiteboard and markers	01
5.	Flip chart stand with paper	2
6.	Jute fiber samples (assorted grades)	Set (1)
7.	Jute product samples (hessian, sacking, CBC, yarn, geo-textile, handicraft)	Set (1)
8.	Carding machine model/working demo unit (or video demonstration setup)	1
9.	Ring spinning frame model/demo unit (or video demonstration)	1
10.	Spinning frame model	1
11.	Winding (Roll winding, Cop winding) (Visual Demonstrations)	1
12.	Dry Beaming (Visual Demonstrations)	1
13.	Dressing (Visual Demonstrations)	1
14.	Weaving	1
15.	Opening Machine (Visual Demonstrations)	1
16.	Damping Machine (Visual Demonstrations)	1
17.	Calendaring (Visual Demonstrations)	1
18.	Mangling (Visual Demonstrations)	1
19.	Lapping (Visual Demonstrations)	1
20.	Cutting (Visual Demonstrations)	1

21.	Sewing (Visual Demonstrations)	1
22.	Baling (Visual Demonstrations)	1
23.	Dust shaker (Visual Demonstrations)	1
24.	Hand loom (for demonstration of weaving principles)	2
25.	Power loom	1
26.	GSM balance (Quadrant balance)	2
27.	Moister meter	5
28.	Digital Weighing scale	1
29.	Tensile strength tester	5
30.	Measuring tape	15
31.	Calculator	5
32.	Techo meter	2
33.	Hackling machine	5
34.	Emulsion tank	1
35.	Softener machine	1
36.	Spreader machine	1
37.	Breaker card (Hand feed, Roll feed)	2
38.	Finisher card	4
39.	Drawing (first, second and third)	3
40.	Trolly	5
41.	Dollop weight	2
42.	Jute table	2
43.	Pile cover	5
44.	Reserve tank	1
45.	Dunnage	10
46.	Digital scale (40 kg)	1
47.	Moister meter	5
48.	Stop watch (digital)	25
49.	First aid kit	1
50.	Fire extinguisher (for safety demonstration)	1

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on **Industry Supervisor (Jute sector)** the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 30 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 30 trainees.